

SPOOFING THE GPS ON SATELLITE PHONES

It is believed that Pakistani intelligence agencies are assisting the terrorists in Kashmir by 'spoofing' the signals of Thuraya satellite phones that the terrorists use, so as to mislead Indian intelligence agencies about their actual location. This article examines whether this is possible or not, and if possible, what countermeasures can be taken to detect or thwart spoofing



■ DR N.C. ASTHANA

The Indian media, citing government sources, claims that Pakistani intelligence agencies are assisting the terrorists in Kashmir by 'spoofing' the signals of the Thuraya satellite phones that the terrorists use. Since intelligence agencies supposedly monitor the terrorists' use of satellite phones, it is believed that the purpose of such spoofing is to make it impossible for the Indian intelligence agencies

to track the location of such satellite phones by making the in-built GPS of the satellite phone give out deliberately wrong coordinates of the handset. Should it be correct, this is a very serious matter because it will be difficult to interdict such terrorists in the first place and also Pakistan can easily deny that its soil is being used for assisting the terrorists.

Since spoofing is a complex subject and little is known about it even amongst most communication engi-

neering professionals, here I present a concise overview of the subject.

Spoofing: more serious than jamming

The popular belief and media perception is that powerful transmitters have been set up across the Line of Control and the international border dividing India and Pakistan, blanking out signals from the Thuraya phones used by terrorists. This is not correct. The strength of the GPS signal on the earth's surface averages -160 dBW. While many GPS receivers leave large space for signal dynamics, enough power space is left for the GPS signals to be overridden. One does not require powerful transmitters to be set up by a nation for that—even low-power transmitters would do the job.

Spoofing is completely different from jamming. A spoof is defined as a malicious signal that overpowers the authentic signal and misleads the receiver to use a forged signal for further processing. Spoofing could be done to gain access to services that are restricted to a certain geographic area (such as a particular game broadcast) or to hide the location of a shipment that is being remotely tracked. It could also be used by an employee to cheat his employer of his true location or by a criminal who is being tracked by the law and does not want the tracker to